

Malasimplex, a simple adversarial perturbation for audio deepfake detectors

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Abstract—The most effective audio deepfake detectors in the literature rely on supervised learning, leveraging public datasets of synthetic speech and data augmentation techniques. While these approaches achieve strong performance on known deepfakes, their ability to generalise to out-of-distribution deepfakes remains challenging. To better understand the frequency signatures exploited by these detectors, we introduce Malasimplex, an adversarial perturbation based on a convolutional filter. We evaluate the Malasimplex perturbation against several SSL-based detectors under scenarios with varying levels of attacker knowledge. Our results show at least a two to three-times increase in Equal Error Rate (EER) in the least effective case, and by up to a six-times increase in the most effective case. Such vulnerability in SSL-based detectors suggests overfitting to specific frequency bands. This work motivates the use of Malasimplex as an interpretability tool for analysing SSL-based audio deepfake detectors, opening a direction for future research.

Index Terms—audio deepfake detection, adversarial perturbation, convolutional filter, self-supervised learning

I. INTRODUCTION

Audio deepfakes are audio recordings generated or manipulated using AI with the aim of deceiving human perception. When used with malicious intent, deepfakes may pose serious risks, for instance in disinformation campaigns or identity fraud. To address this threat, countermeasures (CMs), including Audio Deepfake Detectors (ADDs), have been developed to automatically distinguish manipulated speech from authentic recordings. Evaluation campaigns are conducted to assess the performance of ADDs [1], [2].

Manipulated audio recordings are commonly referred to as spoofs, while genuine recordings are called bona fide. Traditional approaches to spoofing involve generating speech using voice conversion (VC) [3]–[10] or text-to-speech (TTS) [11]–[17]. Recent TTS can generate highly realistic spoofed speech from only a few seconds of audio and support a wide range of languages [11], [12], [15]. Recent VC systems enable zero-shot using speaker-independent representations [3]–[5], [8] or one-shot using a single reference utterance [6]–[8]. Some also support real-time conversion [8]–[10].

In this work, we focus on ADDs leveraging a pre-trained self-supervised learning (SSL) backbone as the frontend and a binary classifier as the backend. SSL-based ADDs are among the best-performing on standard audio deepfake detection benchmarks [18]–[21]. However, their supervised framework

leads to a lack of generalisation to out-of-domain data, where spoofs markers differ from those seen during training. As new spoofs continue to emerge, maintaining robust detection performance remains challenging.

Anomaly detection is a potential alternative to improve robustness to out-of-distribution spoofs. Kim et al. [22] proposed a one-class learning approach but still rely on supervised training. A better understanding of how ADDs make their decisions may help develop new anomaly detection approaches. Recent works have investigated which markers ADDs rely on, in the time domain [23], in the time–frequency domain [24], as well as in the frequency domain with a sub-band analysis [19].

Towards a better understanding of ADDs, adversarial perturbations are an interesting direction. These perturbations are crafted using the knowledge of a target ADD to cause misclassification. We hypothesise that if an adversarial perturbation evades an ADD, the modified parts of the signal are likely to overlap with markers used by ADDs to distinguish authentic from manipulated speech. Furthermore, an interpretable perturbation could provide information about both the spectral signatures of spoofs and the decision-making process of ADDs. This is a long-term objective that motivates Malasimplex but that remains outside the scope of this paper.

Spectral information may help to detect synthetic speech, as shown in [25] and [26]. Their work provide evidence that spoofing markers are unevenly distributed across the spectrum depending on the spoofing algorithms.

To this end, following Malafide [27] and Malacopula [28], we adopt a frequency-domain approach. We propose Malasimplex, an adversarial perturbation that brings spoofs closer to bona fide [29]. Malasimplex is based on a bona fide centroid computed as the mean of bona fide embeddings in a projection space. This projection subspace is adversarially learnt by exploiting the attacker’s knowledge of the target ADD. Malasimplex relies on the strong assumption that an average embedding is a suitable prototype for the bona fide class.

We provide preliminary evidence on the relevance of the identified frequency bands based on their effect on the equal error rate. The focus of this work is, however, to evaluate the effectiveness of these adversarial perturbations by setting up attack scenarios.

The remainder of this paper is organised as follows. Section II describes the typical ADD and attacker blocks. Section III presents Malasimplex, a new adversarial perturbation dedicated to deepfake detection. Section IV is dedicated to the experimental protocol. Section V reports and analyses the experimental results.

II. AUDIO DEEPPFAKE DETECTOR AND ATTACKER

ADDs aim to automatically distinguish spoofed speech from authentic recordings. SSL-based ADDs operate on raw audio waveforms and perform classification on top of the representations learnt by a SSL backbone. Pipeline A in Figure 1 illustrates an attack against such ADDs. Attackers leverage spoofs that resemble bona fide to be misclassified by the target ADD. The adversarial approach is illustrated by pipeline B in Figure 1. Using knowledge of the target ADD, attackers apply perturbations to spoofs to further decrease their likelihood of detection. Focusing on the adversarial approach, we implement pipeline B and describe its key points below.

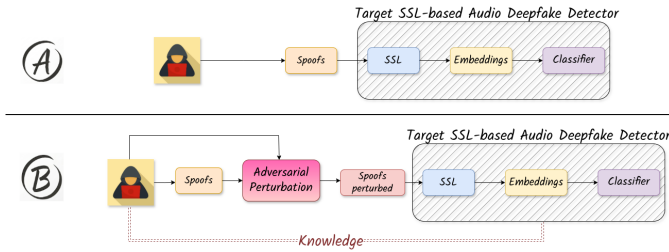


Fig. 1. Overview of traditional spoofing (A) and adversarial perturbation (B) attack pipelines against SSL-based ADDs.

A. Audio Deepfake Detector (ADD)

Current ADDs are modelled as binary classification heads. Top performing ADDs are SSL-based [18]–[21]. In prior works, authors rely on classification heads based on standard pooling strategies, such as global [30] or attentive statistics [22] pooling, as well as more complex architectures such as graph-based pooling with AASIST [27].

ASVSpooof 2019 [31] and 2021 [32] are standard speaker-spoofing training corpora. ASVSpooof 5 [1] corresponds to the latest edition of the challenge. Other corpora gather recent deepfakes (MLAAD [33], LlamaPartialSpooof [34]) or better reflect real-world acoustic conditions (In the Wild [35]).

B. Attacker

To enhance spoofed speech generated by TTS or VC systems in order to evade a target ADD, attackers craft adversarial perturbations. In audio, adversarial perturbations were originally proposed for environmental sound classification and speech command recognition [36], and then extended to target ASV systems [27], [28], [37].

Beyond the effectiveness of these perturbations, key considerations are their perceptual quality and generalisation ability. For instance, some works aim at cross-speaker generalisation [37], whereas other focus on transferability across CMs [27].

In the field of ADD, these perturbations take the form of additive noise or convolutional filters [27], [28], [30], [37]. By exploiting knowledge of the target ADD, these perturbations aim to shift the spoofs in the feature space across the decision boundary, causing misclassification.

The linear perturbation Malafide is optimised on raw utterances based on the target ADD response [27]. Malafide attenuates or amplifies specific frequency bands, with the objective to cause spoofed audio to be misclassified as bona fide. Only spoofed audio is used during training, and no bona fide speech is exploited. Malacopula follows a similar principle but using non linear convolutional filters with embeddings extracted by pre-trained models [28]. The optimisation objective relies on the cosine similarity between embeddings of both bona fide for a given speaker and perturbed spoofs.

However, Malafide and Malacopula show several limitations. First, Malacopula involves a complex process using multiple embedding extractors, with independent filters trained for each pair of speaker and spoofing algorithm. It makes Malacopula difficult to implement and the speaker-dependency is not relevant to the deepfake detection task. Malafide requires knowledge of target ADD’s output scores, a strong assumption in practice.

III. A NEW PERTURBATION: MALASIMPLEX

This section is dedicated to the new perturbation we propose in this article: Malasimplex. It is partly inspired by the centroid one-class method. The centroid one-class approach uses an adaptative centroid for the bona fide class to move the representations of spoofed samples away from it [22].

Malasimplex is designed to optimise a convolutional filter that brings spoofed audio closer to the centroid of bona fide recordings, without any speaker-dependent assumption. Its optimisation is performed in a projection subspace derived from ADD embeddings. Its training procedure consists of two consecutive steps.

A. Step ① : Projection function

The projection function maps audio inputs into a vector representation space built using a dedicated ADD and in which embeddings of spoofed samples are clearly discriminated from those of bona fide samples. The heart of the projection function is a binary classification head trained on embeddings. Let f_{extract} define the encoder, u_i an utterance. f_{extract} maps u_i to its latent representation $f_{\text{extract}}(u_i)$.

The classifier is optimised to maximise inter-class separation. The projection space resulting from Step ① is shown on Figure 2.

B. Step ② : Adversarial filter

Let $\mathbf{h} \in \mathbb{R}^L$ denote the linear time-invariant FIR filter, where L is the number of filter coefficients. The filter is applied in the time domain. A filtered utterance is given by :

$$\text{MS}(u) = u * \mathbf{h}.$$



Fig. 2. Schematics of the projection space optimised given knowledge of the target ADD family.

Let $\mathcal{X}$ denote the current batch. $\mathcal{X}_{\text{spoofs}}$ refers to the subset of spoofed utterances and $\mathcal{X}_{\text{bona}}$ refers to the subset of bona fide utterances, with $\mathcal{X} = \mathcal{X}_{\text{spoofs}} \cup \mathcal{X}_{\text{bona}}$. Each utterance $u_i \in \mathcal{X}_{\text{spoofs}}$ is filtered to get $\text{MS}(u_i)$. Then, $\text{MS}(u_i)$ is mapped to the projection space learnt in Step ① to get the embedding of the filtered utterance :

$$z_i^{\text{pert}} = f_{\text{extract}}(\text{MS}(u_i)).$$

Next, the centroid of bona fide embeddings is computed as the average representation of the bona fide embeddings :

$$\hat{z}_{\text{bona}} = \frac{1}{N_{\text{bona}}} \sum_{i=1}^{N_{\text{bona}}} f_{\text{extract}}(u_i), \quad \forall u_i \in \mathcal{X}_{\text{bona}}$$

The filter $\mathbf{h}$ is optimised to minimise the cosine loss $\mathcal{L}$ between the perturbed spoofs embeddings and the bona fide centroid :

$$\mathcal{L} = 1 - \frac{1}{N} \sum_{i=1}^N \cos(\hat{z}_{\text{bona}}, z_i^{\text{pert}})$$

Filter weights are updated according this loss until convergence. Step ② results in a representation space where perturbed spoofs and bona fide samples are closer than initially, as illustrated on the right side of Figure 2.

We hypothesise that the centroid may be a suitable prototype for the bona fide class, even if there is no guarantee that this averaged representation has a physical meaning. Step ① favours inter-class separation but does not take into account intra-class compactness.

IV. EVALUATION PROTOCOL

This section introduces the evaluation protocol. Our first objective is to verify that the Malasimplex filter is capable of compromising an ADD. The next objective is to assess the generalisation of Malasimplex depending on the attackers knowledge of the target ADD. In particular, we detail different scenarios, from a white-box to a black-box scenario, depending on the amount of knowledge of the target ADD. The assumption is that the more the attackers know about their target, the higher is their ability to evade the target. However, we want to ensure that the proposed approach remains effective even under unfavourable conditions for the attacker, despite an expected performance degradation.

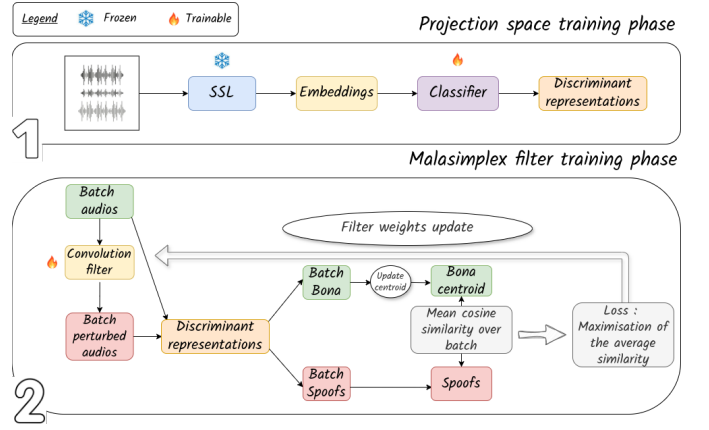


Fig. 3. Schematics of the adversarial model Malasimplex in two phases.

A. Target ADD

In this section, several ADDs are defined by varying their architecture, including their backbone and classification head. They serve as target ADDs in the evaluation section for evaluating Malasimplex in different attack scenarios.

They are trained using widely used benchmark corpora : ASVSpoo5, M-aillabs [38], MLAAD, ASVSpoo2019 and ASVSpoo2021. We apply similar data augmentation than effective ADDs in the literature [39]–[41].

We build target ADDs based on two different SSLs : Wav2Vec XLS-R 300M [42], [43] and WavLM Large [44]. We select as the backend the Multi-Head Factorized Attention (MHFA) [45] and Conformer [46] heads. For both heads, the latent space size is set to 1024. The backbone is kept frozen, only classification head weights are updated during training. These architectures are among the best performing during ASVSpoo5 challenge [47], [48].

The standard evaluation metric used in the ASVSpoo5 challenges is the equal error rate (EER).

EERs achieved by the target ADDs across different configurations on standard corpora are reported in Table I. EERs are computed on the official evaluation lists when available, and on the entire corpus otherwise. For LlamaPartialSpoo, the EER is computed only on the subset of fully spoofed utterances.

In-domain corpora use the official training partition (or a subset when unavailable) to train the target ADD. Out-of-domain corpora exhibit acoustic characteristics that differ from the training data.

On out-of-domain datasets, the XLS-R-based ADDs show the best performance with EERs ranging from 2% to 4%. The WavLM-based ADD achieve higher EERs, between 4 and 6%. These results show that the training protocols allow us to reach performance levels comparable to top-ranked models on the SpeechArena DF leaderboard [18].

B. Spoofs

For training the projection function of our Malasimplex perturbations (subsection III-A), we use audio samples drawn from ASVSpoo5-Eval. Each projection function is trained using a single TTS system. We select YourTTS [11] and XTTS

Target ADD	ASV5 Eval	ASV2019 LA	ASV2021	MLAAD-Mailabs	In the Wild	LlamaPartial Spoof
	In-domain				Out-of-domain	
XLS-R	6.98	1.21	2.30	0.04	1.70	3.75
XLS-R conformer	9.75	0.80	1.64	0.19	2.20	2.54
WavLM	6.38	1.53	4.12	0.08	5.93	4.83

TABLE I
EER (%) OF TARGET ADDS BASED ON TWO SSL ENCODERS.

[15] as they are high quality and popular TTS [49]. XTTS spoofs are seen by target ADDs since they belong to MLaAD, whereas YourTTS spoofs are never seen during their training.

For the inference step, the perturbation is kept frozen. It can thus be applied to spoofs generated by TTS unseen during Malasimplex training. We use in each evaluation setting either audio samples from a single TTS system or a combination of synthetic audio sources.

- The single TTS systems are YourTTS and XTTS.
- The combination consist of the In the Wild corpus, whose spoofing algorithms are unknown.

C. Malasimplex filters training

This subsection provides details on Malasimplex filters training, referring to Step ② in Figure 3. The first filter denoted MS 1 is trained as follows: the projection function is built on a Wav2Vec2 XLS-R backbone. MS 1 is trained to separate bona fide utterances from YourTTS spoofs. Other hyperparameters are fixed and summarised in Table II.

Hyperparameter	Value
Training data	ASVSpooF5-Eval
<i>Step ① – Projection function learning</i>	
Classification head	MHFA
Number of attention heads	8
Latent dimension	128
Data augmentation	Same as target ADDs
Batch size	128
<i>Step ② – Filter training</i>	
Filter length L	257
Batch size	64

TABLE II
FIXED HYPERPARAMETERS USED TO TRAIN ALL MALASIMPLEX FILTERS.

In our study, we train several Malasimplex filters as described previously, by varying the SSL backbone and the spoofs on which the projection function is trained. Table III summarises the filters configurations.

The bona fide samples not used to train the projection function are set aside to construct the evaluation sets. Each evaluation set combines these bona fide samples and a corresponding set of spoofed samples (subsection IV-B). The exception is the In the Wild corpus which already contains bona fide samples and is used unchanged.

Name	SSL Backbone	Training set of projection function
MS 1	Wav2Vec XLS-R	YourTTS
MS 2	WavLM Large	YourTTS
MS 3	WavLM Large	XTTS

TABLE III
CONFIGURATIONS OF TRAINED MALASIMPLEX FILTERS

D. Scenarios

Several scenarios are defined according to the spoofing data setting and the attacker’s level of knowledge of the target ADD, as summarised in Table IV. All scenarios share a common characteristic: the attacker is assumed to know the data augmentation protocol used by the defender, as it is a standard protocol [39]–[41]. None of the scenarios requires knowledge of the weights or gradients of the target ADD.

Scenario	Spoof seen by target	Attacker knowledge of target	
		Type of classif. head	Backbone
White-box	✗	✓	✓
Gray-box	✗	✓	✗
Gray-box in-domain	✓	✓	✗
Black-box	✗	✗	✗

TABLE IV
DIFFERENT ATTACKER SCENARIOS.

The first scenario is referred to as white-box, following common usage in the adversarial literature, although only the architecture of the target ADD is known here. It simulates the most favourable case among the scenarios considered, as the attacker can use the same backbone and classification head architectures as the target ADD to train a surrogate ADD.

The next scenario is a gray-box one, where the attacker has only a partial knowledge of the target ADD. Partial knowledge includes the architecture of the classification head, e.g. MHFA head. This scenario is designed to evaluate the generalisation of Malasimplex in cases where the SSL backbone of the target ADD differs from that used by the attacker for training the projection function.

The first two scenarios assume the attacker can generate synthetic speech using spoofing algorithms unseen by the target ADD during training. This can be known directly, from the target ADD training data, or indirectly, from its outputs, by selecting the spoofs least well detected. We assume this favours the attacker, since such spoofs are less detectable regardless of any adversarial perturbation. The scenario named gray-box in-domain removes this knowledge: the attacker no longer knows which algorithms were excluded from training, nor can they access the target ADD outputs to identify weakly-detected spoofs. This allows us to assess Malasimplex generalisation under less favourable conditions. It is referred to as in-domain because the attacker uses spoofs possibly seen during training, and thus more easily detected on their own.

In the black-box scenario, the target ADD classification head is unknown to the attacker. This represents another difficulty for the attacker compared to the previous scenarios.

In each scenario, the perturbation is compared to a randomly-initialised convolution filter to ensure that the observed effect is not merely due to artifacts introduced by the convolution operation and frame overlap. Random filters are drawn using Kaiming uniform initialisation.

V. RESULTS

This section details the experimental results, in which Malasimplex perturbations (Table III) are used to attack several target ADDs across different scenarios.

A. Malasimplex across projection spaces

In this experiment, we consider a gray-box scenario where the attacker knows the classification head of the target but not its SSL backbone, and chooses spoofs that have not been seen during target training (IV). The goal is to assess Malasimplex generalisation to a projection space derived from a SSL backbone other than the one on which it was trained.

Results are reported in rows ② and ③ part of Table V. MS 1 generalises well across other SSL backbones as it succeeds to increase the initial EER by an average of 2 to 3 times on WavLM Large backbone (for instance on In the Wild, from 5.91 to 15.09% and from 1.33 to 3.83% on XTTS). MS 2 generalises as well on XLS-R backbone as it multiplies the EER by an average of 3 to 5 (1.70 to 8.47% on In the Wild and 0.99 to 4.35% on XTTS).

Both filters are trained exclusively on YourTTS spoofs. They perform similarly on YourTTS, XTTS, and In the Wild, although the standard deviation on In the Wild is higher. The relative EER increase achieved by a given filter depends on the evaluation set. For instance, for MS 2, the EER increases by a factor of 5 on In the Wild, compared with around 3 on YourTTS-Eval.

B. Malasimplex in non-optimal conditions

The gray-box in-domain scenario adds an additional difficulty compared to the previous scenario : the attacker chooses spoofs that have been seen during target training. Therefore, they are more likely to be well detected by the target. It assesses the effect of the chosen training spoofs on Malasimplex performance. Results are reported in row ④ part of Table V. On XTTS-Eval, MS 3 underperforms MS 2. Instead of increasing the initial EER by a factor 4 on average as in the row ③ (from 0.99 to 4.35%), MS 2 increases it by only a factor 2 on average (from 0.99 to 1.57%).

VI. ABLATIONS

This section presents an ablation study. In particular, we analyse the effect of fine-tuning the SSL encoder and the choice of the classification head on Malasimplex performance.

A. Fine-tune or not the SSL ?

The white-box scenario enables to assess Malasimplex difference of performance depending on the backbone fine-tuning strategy used. Results are reported in row ① part of Table V. MS 1 FT denotes the same configuration as MS

1 (Table III), but with the SSL backbone fine-tuned during projection function learning (Figure 3).

The filter MS 1 FT performs nearly twice as well as MS 1. Indeed, on YourTTS spoofs, the metric increases by 3 times on average for MS 1 FT (from 3.26 to 10.86%) compared to twice on average for MS 1 (from 3.26 to 7.60%). However, on YourTTS spoofs, the PESQ score is 3.20 ± 0.25 under MS 1 and 2.25 ± 0.30 under MS 1 FT perturbation. MS 1 FT significantly reduces audio quality compared to MS 1.

B. Malasimplex across classification head architectures

The black-box results are reported in row ⑤ of Table V. The target ADD built with the Conformer head has similar initial performance to that built with the MHFA head. Here, random perturbations have a stronger effect, increasing the initial EER by a factor of 1.5 to 2 on average, depending on the SSL and spoof set. For instance, the EER increases from 2.14 to 10.00% on In the Wild and from 3.66 to 15.9% on YourTTS spoofs.

However, comparing the EERs relative to the random perturbation reveals the increase in EER is slightly smaller than in the gray-box scenario, increasing by a factor of 2 to 4 after applying Malasimplex. The absolute increases in EER are much more pronounced than the relative changes. The absolute increase is around 10 percentage points on all sets.

VII. DISCUSSION

The results first suggest that Malasimplex is transferable across different SSL representations. This implies that an attacker does not need to know the SSL used by the target ADD to build an effective perturbation.

Second, the results indicate that Malasimplex perturbations are also transferable across different spoofing algorithms. We observed it on both YourTTS and XTTS but also a corpus containing unknown deepfakes. An attacker could therefore use a spoofing algorithm that is least detectable by the target ADD and then apply the resulting perturbation to other spoofs.

Thus, Malasimplex perturbations appear to be agnostic to both the underlying SSL representations, when these representations are sufficiently informative for the original task, and to the spoofing algorithms, when the resulting spoofs are less detectable by the target ADD.

However, these observations should be considered in light of the fact that both SSLs considered here are among the most effective for deepfake detection. They share characteristics in both their architectures and training data. Despite relying on different pre-training strategies, contrastive for Wav2Vec and predictive for WavLM, they might converge to similar representation subspaces [50]. Even when frozen, their representations remain informative for the task, which may explain the observed cross-SSL generalisation.

Indeed, we evaluated Malasimplex using frozen backbones as proposed in [51]. This raises the question of the role of the fine-tuning on Malasimplex effectiveness. To answer this question, we did an ablation study. We observed that the

Row	Scenario	Target ADD	Evaluation set	Without perturbation ↓	With random perturbation	With adv. perturbation ↑	Perturbation
①	White-box	XLS-R MHFA	YourTTS-Eval	3.26	3.22 ± 0.38	7.60 ± 1.23 10.86 ± 2.05	MS 1 MS 1 FT
②	Gray-box	WavLM MHFA	YourTTS-Eval XTTS-Eval In the Wild	5.10 1.32 5.94	3.93 ± 0.46 1.02 ± 0.19 5.91 ± 0.88	11.82 ± 1.12 3.83 ± 0.64 15.09 ± 2.16	MS 1 <i>XLS-R YourTTS</i>
③	Gray-box	XLS-R MHFA	YourTTS-Eval XTTS-Eval In the Wild	3.27 0.99 1.70	3.22 ± 0.38 0.88 ± 0.15 2.37 ± 0.27	8.81 ± 1.79 4.35 ± 1.97 8.47 ± 2.39	MS 2 <i>WavLM YourTTS</i>
④	Gray-box in domain	XLS-R MHFA	XTTS-Eval	0.99	0.88 ± 0.15	1.57 ± 0.15	MS 3 <i>WavLM XTTS</i>
⑤	Black-box	XLS-R Conformer	YourTTS-Eval XTTS-Eval In the Wild	3.66 1.60 2.14	7.07 ± 0.52 3.22 ± 0.28 3.60 ± 0.44	15.9 ± 2.78 10.03 ± 3.65 10.00 ± 1.11	MS 2 <i>WavLM YourTTS</i>

TABLE V
EER (%) (MEAN ± STD OVER 10 SEEDS) OF TARGET ADDS ON EVALUATION SETS.
 RESULTS ARE REPORTED WITHOUT PERTURBATION, WITH A RANDOM PERTURBATION AND THEN WITH MALASIMPLEX PERTURBATION.
 THE COLUMN WITHOUT PERTURBATION ↓ CORRESPONDS TO THE DEFENDER’S PERSPECTIVE.
 THE COLUMN WITH ADVERSARIAL PERTURBATION ↑ CORRESPONDS TO THE ATTACKER’S PERSPECTIVE.

improvement in attack performance achieved through fine-tuning is significant from an attacker’s perspective, but leads to a reduction in the perceptual quality of the filtered audio.

The results also highlight that if a target ADD has seen specific spoofs, Malasimplex drops significantly in performance when the attacker select the same spoofs. This is a finding consistent with [30].

Since the choice of spoofs during MS training is critical, we further investigated whether knowledge of the target classification head is also critical. Unexpectedly, the ablation study shows that using a different classification head leads to a larger absolute increase in error rates. Random perturbations are even more effective at evading the target ADD in this setting. This suggests that the Conformer head may overfit to frequency markers. This finding reveals a weakness of ADDs and highlights the importance of selecting relevant information in attention layers, as done by the MHFA head.

Additional material, including audio samples and filter visualisations with multiple seeds, is available here.

CONCLUSION

In this study, we adopt an adversarial approach based on the assumption that an effective perturbation reveals attacks markers exploited by Audio Deepfake Detectors (ADDs). To this end, as a continuation of the previously proposed Malafide and Malacopula perturbations, we introduced Malasimplex.

Malasimplex differs by its simplification for the deepfakes detection task as its optimisation does not require speaker-dependency. Especially, it relies on an average of embeddings to characterise the bona fide class.

We assume that the frequency bands highlighted by Malasimplex can be used to improve our understanding of both the decision-making process and the vulnerabilities of

ADDs. This work motivates the development of frequency-based anomaly detection methods.

The results demonstrate that Malasimplex perturbations allow an attacker to significantly degrade a SSL-based target ADD performance across several scenarios. The partial knowledge of the target ADD allows Malasimplex perturbations to a two to three-times increase in error rates in the least effective case, and by up to six times in the most effective case. This supports our initial hypothesis that SSL-based ADDs rely on specific, potentially correlated, frequency bands for classification. It therefore suggests that frequency-domain anomalies constitute an encouraging direction.

Further studies should validate the use of Malasimplex as an interpretable tool for understanding ADDs, especially by investigating the frequency bands significance. One way to assess the frequency bands significance is investigating the effect of the perturbation on bona fide speech. We expect from interpretable frequency bands that the perturbation should be ineffective on it. We could also use clean speech to isolate the influence of an individual controlled signal distortion such as codecs, noise or reverberation, on the frequency bands.

Supporting this perspective of studying stability, we observed that even random perturbations can increase the error rates by up to two to three times in a specific setting. It is possible that others random perturbations, with different initialisations or frequency ranges, could achieve higher effectiveness. Therefore, it remains to be determined to what extent the changes Malasimplex induces in the latent space translate into meaningful modifications in the frequency domain, and to what extent it takes advantage of unstable decision boundaries.

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AI-GENERATED CONTENT DISCLOSURE

ChatGPT (OpenAI) and Claude (Anthropic) were used to support the improvement of the manuscript, particularly for reformulation and writing style enhancement.

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